

Events of a translating edge: lossless code length and rate under a timing tolerance

event-coding project, theory note 1 (revised after independent checks)

October 9, 2026

Summary. An edge that translates across M event pixels produces M copies of one event pattern. Given the edge and its velocity, the copies differ only by a threshold phase per pixel and a timing jitter per event. Theorem 1 gives the lossless code length, which stays constant per event as M grows. Theorem 2 gives the rate under a mean-squared timing error D : it is known exactly for $D \leq \sigma^2$, within 0.255 bit per pixel above that, and it is zero from $D_A = \sigma^2 + \tau_\varphi^2/12$ on. Corollary 1 shows that beyond D_A a pass over M pixels needs at most $2 \ln M + O(1)$ nats in total. Rates are in nats unless bits are named.

1 Model

Pixels sit at integer positions $i = 0, \dots, M - 1$ on a line. A continuous log-intensity profile g translates at speed $v > 0$ pixels per second, so pixel i observes

$$s_i(t) = g(vt - i). \quad (1)$$

(A1) Pixel. Pixel i holds a reference r_i . It emits an event at the first time t at which $|s_i(t) - r_i| = C$, with polarity $p = \text{sign}(s_i(t) - r_i)$, and then sets $r_i \leftarrow r_i + pC$. The threshold $C > 0$ is common to all pixels and known to the decoder.

(A2) Edge. $g(u) = 0$ for $u \leq 0$, $g(u) = \gamma u$ for $0 \leq u \leq w$, and $g(u) = V$ for $u \geq w$, with $\gamma > 0$ and $V = \gamma w = nC$ for an integer $n \geq 1$.

(A3) Phase, first pass. Before the edge arrives, $r_i(0) \in (-C, 0]$. The phases $a_i = r_i(0) + C \in (0, C]$ are independent, uniform, and independent of everything else.

(A4) Jitter and clock. The k th event of pixel i is recorded at $T_{i,k} = t_{i,k} + \varepsilon_{i,k}$, where $t_{i,k}$ is the time given by (A1) and the $\varepsilon_{i,k}$ are independent $\mathcal{N}(0, \sigma^2)$, independent of the phases, with $\sigma > 0$ the same for all events. The stored value is $Y_{i,k} = \delta \lfloor T_{i,k}/\delta \rfloor$. The level index k of each recorded event is known.

Write $\tau_\varphi = C/(\gamma v)$ for the time between successive events of one pixel, and $\eta(a, \sigma) = h(U_a + Z_\sigma)$ for the differential entropy of the sum of independent $U_a \sim \text{Unif}(0, a)$ and $Z_\sigma \sim \mathcal{N}(0, \sigma^2)$. Then

$$\frac{1}{2} \ln(a^2 + 2\pi e \sigma^2) \leq \eta(a, \sigma) \leq \frac{1}{2} \ln(2\pi e(a^2/12 + \sigma^2)), \quad (2)$$

by the entropy power inequality and by the Gaussian maximum-entropy bound at the variance of $U_a + Z_\sigma$. Moreover $\eta(a, \sigma) - \ln a \rightarrow 0$ as $\sigma/a \rightarrow 0$.

2 Structure of the events

Lemma 1. *Under (A1), every event of pixel i occurs at a time t with $s_i(t) \in a_i + C\mathbb{Z}$. Under (A1) to (A3), pixel i emits exactly n events, all of positive polarity, at*

$$t_{i,k} = \frac{i}{v} + A_i + (k-1)\tau_\varphi, \quad k = 1, \dots, n, \quad A_i = \frac{a_i}{\gamma v} \sim \text{Unif}(0, \tau_\varphi]. \quad (3)$$

Proof. The reference starts in $a_i + C\mathbb{Z}$ and changes by $\pm C$, so it stays there, and an event requires $s_i = r_i \pm C$. Under (A2), s_i is nondecreasing and $r_i(0) \leq 0 = s_i(0)$, so events occur when s_i reaches $a_i + (k-1)C$, $k = 1, 2, \dots$, as long as this level is at most $V = nC$. Since $0 < a_i \leq C$, exactly the levels $k = 1, \dots, n$ are reached. Solving $\gamma(vt - i) = a_i + (k-1)C$ gives (3). $\square$

In the coordinate $u = vt - i$ attached to the profile, an event is a point of the graph of g on the level lattice $a_i + C\mathbb{Z}$ of its pixel. Given (g, v) , the noise-free events of pixel i are a function of the single number a_i .

3 Lossless code length

Let $T_i = (T_{i,1}, \dots, T_{i,n})$ and $Y_i = (Y_{i,1}, \dots, Y_{i,n})$.

Theorem 1. *Under (A1) to (A4), the vectors $T_0, \dots, T_{M-1}$ are independent given (g, v) ,*

$$h(T_i) = \frac{n-1}{2} \ln(2\pi e \sigma^2) + \eta(\sqrt{n} \tau_\varphi, \sigma), \quad (4)$$

and, as $\delta/\sigma \rightarrow 0$,

$$H(Y_i | g, v) = n \ln \frac{1}{\delta} + \frac{n-1}{2} \ln(2\pi e \sigma^2) + \eta(\sqrt{n} \tau_\varphi, \sigma) + o(1). \quad (5)$$

Proof. By Lemma 1, $T_i = c_i + A_i \mathbf{1} + \varepsilon_i$ with c_i a function of (g, v) , $\mathbf{1}$ the all-ones vector, and $\varepsilon_i \sim \mathcal{N}(0, \sigma^2 I_n)$. Independence over i follows from (A3) and (A4). Let Q be an orthogonal matrix whose first row is $\mathbf{1}^\top / \sqrt{n}$. Then $Q(T_i - c_i) = (X_i, W_i)$ with

$$X_i = \sqrt{n} A_i + \varepsilon_i^\parallel, \quad W_i \sim \mathcal{N}(0, \sigma^2 I_{n-1}), \quad (6)$$

where $\varepsilon_i^\parallel \sim \mathcal{N}(0, \sigma^2)$, and X_i and W_i are independent because $Q\varepsilon_i$ has independent coordinates and A_i is independent of ε_i . Differential entropy is invariant under translations and orthogonal maps, so $h(T_i) = h(X_i) + h(W_i)$, which is (4). Since $\sigma > 0$, T_i has a bounded density and finite variance, and (5) is the relation between the entropy of a uniformly quantized vector and its differential entropy [1, Ch. 8]. $\square$

When $\sigma \ll \tau_\varphi$ the cost per event is

$$\frac{H(Y_i | g, v)}{n} \approx \frac{n-1}{n} \ln \frac{\sigma \sqrt{2\pi e}}{\delta} + \frac{1}{n} \ln \frac{\sqrt{n} \tau_\varphi}{\delta}. \quad (7)$$

One event per pixel pays for the phase and the other $n-1$ pay for jitter only. Neither term depends on M : a decoder that knows the edge and its velocity exactly still pays (7) for every event. With $\sigma = 0.3$ ms, $\tau_\varphi = 2$ ms, $n = 3$ and $\delta = 1 \mu\text{s}$ this is 10.8 bits per event.

4 Rate under a timing tolerance

The encoder and the decoder know (g, v) . The decoder outputs $\hat{T}_i \in \mathbb{R}^n$ for each pixel, and the distortion is $d(T_i, \hat{T}_i) = \frac{1}{n} \|T_i - \hat{T}_i\|^2$, the mean-squared error against the recorded times. Let $R_1(D)$ be the rate-distortion function of T_i in nats per pixel. It is an operational rate for joint coding of many pixels. Define

$$s_X^2 = \frac{n\tau_\varphi^2}{12} + \sigma^2, \quad D_A = \sigma^2 + \frac{\tau_\varphi^2}{12}, \quad \Gamma = \frac{1}{2} \ln(2\pi e s_X^2) - \eta(\sqrt{n} \tau_\varphi, \sigma). \quad (8)$$

By (2), $0 \leq \Gamma \leq \frac{1}{2} \ln(2\pi e/12)$, which is 0.255 bit, for every σ .

Theorem 2. *Under (A1) to (A4):*

$$R_1(D) = h(T_i) - \frac{n}{2} \ln(2\pi e D), \quad 0 < D \leq \sigma^2, \quad (9)$$

$$[U(D) - \Gamma]^+ \leq R_1(D) \leq U(D), \quad U(D) = \frac{1}{2} \ln \frac{s_X^2}{nD - (n-1)\sigma^2}, \quad \sigma^2 < D < D_A, \quad (10)$$

$$R_1(D) = 0, \quad D \geq D_A, \quad (11)$$

and $R_1(D) > 0$ for $D < D_A$.

Proof. The distortion is invariant under the translation by c_i and the orthogonal map Q , so R_1 is the rate-distortion function of the independent components (X_i, W_i) of (6) under total squared error nD . Their variances are s_X^2 and σ^2 ($n-1$ times).

Lower bound for all D . Let $(\hat{X}, \hat{W})$ have component errors d_X and $d_1, \dots, d_{n-1}$ with $d_X + \sum_j d_j \leq nD$. Independence of the components gives $I(X, W; \hat{X}, \hat{W}) \geq I(X; \hat{X}) + \sum_j I(W_j; \hat{W}_j)$. For a Gaussian component $I(W_j; \hat{W}_j) \geq \frac{1}{2} \ln^+(\sigma^2/d_j)$, and $I(X; \hat{X}) \geq h(X) - h(X - \hat{X}) \geq \eta - \frac{1}{2} \ln(2\pi e d_X)$. The function $\ln^+(\sigma^2/d)$ is convex in d , so the Gaussian terms are smallest when all d_j equal a common d_W , and $d_W > \sigma^2$ brings no benefit. For $n \geq 2$ this gives

$$R_1(D) \geq \min_{\substack{0 < d_W \leq \sigma^2 \\ d_W < nD/(n-1)}} \left[\eta - \frac{1}{2} \ln(2\pi e (nD - (n-1)d_W)) + \frac{n-1}{2} \ln \frac{\sigma^2}{d_W} \right]. \quad (12)$$

The bracket is convex in d_W and its derivative vanishes only at $d_W = D$. The minimum is at $d_W = D$ when $D \leq \sigma^2$, where it equals the right side of (9), and at $d_W = \sigma^2$ otherwise, where it equals $U(D) - \Gamma$. For $n = 1$ there is no W and the same values follow with $d_X = D$.

Equality for $D \leq \sigma^2$. Write $\varepsilon_i = \varepsilon' + N$ with independent $\varepsilon' \sim \mathcal{N}(0, (\sigma^2 - D)I_n)$ and $N \sim \mathcal{N}(0, DI_n)$, and let $\hat{T} = c_i + A_i \mathbf{1} + \varepsilon'$. Then $T_i = \hat{T} + N$ with N independent of $\hat{T}$, the distortion is D , and $I(T_i; \hat{T}) = h(T_i) - h(N)$, which is the lower bound.

Upper bound for $\sigma^2 < D < D_A$. Spend no rate on W_i , at error σ^2 per component, and describe X_i at error $d_X = nD - (n-1)\sigma^2 \in (\sigma^2, s_X^2)$. A source of variance s^2 needs at most $\frac{1}{2} \ln(s^2/d)$ nats at squared error d [1, Ch. 10].

Zero rate. With no description the best output is the mean, at distortion $\frac{1}{n}(s_X^2 + (n-1)\sigma^2) = D_A$. This is the smallest distortion at which a rate-distortion function vanishes [1, Ch. 10]. $\square$

Per event and per decade of the root-mean-square tolerance $\sqrt{D}$, the rate falls by $\log_2 10 = 3.32$ bits for $\sqrt{D} < \sigma$. Above σ the slope of U is

$$\frac{3.32}{n} \left[1 - \frac{(n-1)\sigma^2}{nD} \right]^{-1} \text{ bits}, \quad (13)$$

which decreases continuously from 3.32 at $D = \sigma^2$ to $3.32 D_A/s_X^2$ at D_A , a value close to $3.32/n$ when $\sigma \ll \tau_\varphi$. The rate reaches zero with a kink at D_A . The intermediate regime, in which only one phase per pixel is described, is distinct only when $\sigma \ll \tau_\varphi$. At $D = D_A$ the decoder regenerates the events of every pixel at the mean phase, $\hat{T}_{i,k} = i/v + (k - \frac{1}{2})\tau_\varphi$. Each regenerated event is then within $\tau_\varphi/2$ of its noise-free time, and the count and the polarities are exact.

Corollary 1. *Let (A1) to (A4) hold, let γ and v range over a compact set bounded away from zero and the edge position over a range of order M , let the encoder know these three parameters, and let the decoder know C , σ , n and the pixels traversed. Fix $\beta > 0$. For $D \geq (1 + \beta)D_A$ the nM events of a pass over M pixels can be described with at most*

$$B = 2 \ln M + O(1) \quad \text{nats in total, that is } O\left(\frac{\ln M}{M}\right) \text{ per event}, \quad (14)$$

where the $O(1)$ term depends on the ranges, n , β , C and σ .

Sketch. By (3) the mean-phase times have sensitivity $1/v$ to the position, at most $n\tau_\varphi/\gamma$ to γ , and at most of order M/v^2 to v . Quantize each parameter so that the total induced timing error is at most $\sqrt{\beta D_A}$ for every event. This takes $\ln M + O(1)$ nats for v , the same for a position with range of order M , and $O(1)$ for γ . Given the parameters, the mean-phase error has zero mean and the parameter-induced error is deterministic, so the two are orthogonal and the distortion is at most $(1 + \beta)D_A$. $\square$

The lossless cost of the same events is M times (5), so the ratio of the two grows at least as $M/\ln M$. Above $\sqrt{D_A}$ the description that remains is $O(\ln M)$. Below it a cost $R_1(D) > 0$ per pixel remains.

5 Scope

Remark 1 (The phase is a property of the pixel). Under (A1) the lattice $a_i + C\mathbb{Z}$ never changes. After a rising pass with $V = nC$ the reference sits at $a_i + (n - 1)C$, so a second rising edge meets the same phase a_i . A falling edge first returns through the last level crossed, which does not fire, and yields $n - 1$ events, as does every later pass in either direction. Assumption (A3) therefore describes a first pass with phases set by earlier, unmodeled activity. With repeated passes and nothing else, the phase is paid once per pixel, and the rate per event in the intermediate regime of Theorem 2 falls in proportion to the number of events that share the phase. The persistence relies on the reset $r_i \leftarrow r_i + pC$. A pixel whose reset stores the instantaneous signal, or whose ON and OFF thresholds differ, redraws the phase.

Remark 2 (Departures from (A2) and (A3)). (i) *Firing rule*. For any continuous signal, a lattice level fires when crossed unless it was the last level crossed, with the initial reference counted as a crossed level. A reversal whose two adjacent legs both exceed C skips exactly one level. If $r_i(0) \in (0, C)$, the first level is skipped and the pixel emits $n - 1$ events. (ii) *Non-integer contrast*. If $V/C = n + f$ with $0 < f < 1$, the count is $n + 1$ when $a_i \leq fC$ and n otherwise. Reproducing the count exactly costs the binary entropy of f per pixel, which does not decay with M . Equation (11) and Corollary 1 then hold only for a distortion measure that tolerates one unmatched event per pixel. (iii) *Smooth monotone edge*. The event times are $t_{i,k} = (i + g^{-1}(a_i + (k - 1)C))/v$. The zero-rate distortion becomes $\sigma^2 + \frac{1}{nv^2} \sum_k \text{Var } g^{-1}(a_i + (k - 1)C)$, reached by regenerating at the mean times. For an edge that joins two flat levels the first and last events have the largest terms, because g' vanishes there. The phase then moves T_i along a curve and the decomposition (6) no longer applies. (iv) *Order*. When $\sigma \gtrsim \tau_\varphi$ the recorded order can differ from the level order. The entropy of the unordered set is lower by at most $\ln n!$.

Remark 3 (Relation to sampling theory). Pixel i sees the delayed signal $s_0(t - i/v)$, as one channel of the generalized sampling expansion does [2]. The M lattices give Mn distinct levels with typical spacing of order C/M , the amplitude counterpart of multi-channel time encoding [3]. The profile has few parameters, so this resolution is not needed to regenerate events at tolerance $\sqrt{D_A}$.

Remark 4 (Not covered). Noise events add their own cost: about $1 - \ln(\lambda\delta)$ nats per event for a Poisson stream of rate λ per pixel with $\lambda\delta \ll 1$, plus $\ln 2$ for a random polarity, or nothing if the decoder may drop them. Under (A1) a noise event leaves the lattice intact and changes the count of the next edge by one. Threshold mismatch, refractory time, finite pixel bandwidth, a jitter that depends on the slope, an unknown profile learned from the events, acceleration, and two-dimensional geometry are outside this note. For a straight edge in two dimensions only the velocity component normal to the edge enters.

6 Numerical check

`check_translation_edge.py` tests the statements on synthetic data. (N1) A time-stepping simulation of (A1) reproduces (3) for a ramp, and the level-crossing times of Remark 2(iii) with counts 3 or 4 for a smooth edge with $V/C = 3.2$, to within the time step. The firing rule of Remark 2(i) agrees with (A1) on 20,000 random zigzag signals. (N2) A nearest-neighbor estimate of $h(T_i)$ from 2×10^5 samples agrees with (4) to within 0.035 nat at five pairs $(n, \sigma/\tau_\varphi)$, with n from 2 to 5 and σ/τ_φ from 0.05 to 1.5. (N3) For $n = 3$, $\sigma = 0.1$ ms and $\tau_\varphi = 5$ ms, scalar entropy-coded uniform quantizers give the operating points below, in bits per event. The bounds are those of Theorem 2 for R_1 , which needs joint coding over pixels, so a scalar coder may exceed the upper bound. The tabulated points lie 0.25 to 0.37 bit per described component above the lower bound, close to the 0.255 bit loss of uniform scalar quantization at fine resolution.

coder	$\sqrt{D}$ (ms)	measured rate	lower bound on R_1/n	upper bound on R_1/n
mean-phase regeneration	1.447	0.000	0.000	0.000
phase only	0.337	0.760	0.640	0.715
phase only	0.117	1.388	1.297	1.373
all components	0.029	3.559	3.266	3.266
all components	0.003	6.842	6.588	6.588

Independent reviews of this note also confirmed (4) by direct integration of the density and (9) to (11) by Blahut-Arimoto computations of R_1 , one of them on T_i directly for $n = 2$ without the split (6).

7 Predictions for recorded data

The theorems suggest these under the added conditions named. None is proved. (P1) For in-box events coded by a model that knows the motion, the code length against clock resolution δ leaves the slope of 3.32 bits per decade near $\delta \sim \sigma$, falls gradually, and is flat beyond a few τ_φ . For background events the slope stays 3.32 while $\lambda\delta \ll 1$. This transfers Theorem 2 from mean-squared error to clock resolution and assumes one event spacing per object and few noise events inside the box. A spread of spacings smears the transition.

(P2) A regenerating decoder reaches a root-mean-square timing error near $\sqrt{\sigma^2 + \tau_\varphi^2/12}$ at no cost per pixel on a first pass, for a ramp-like edge. Model error in the profile and the motion is expected to add to it.

(P3) At a tolerance above $\sqrt{D_A}$, bits per event of a regenerating coder fall with the path length at least as fast as $(2 \ln M + c)/(nM)$, where the constant c can exceed $2 \ln M$ at practical path lengths, until one of the following sets a floor: unmatched events from non-integer contrast, new content, occlusion, or non-rigid motion.

References

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